ANKIT RAJENDRA SINGH

Full Stack Engineer — Backend-Focused

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PROFESSIONAL SUMMARY

Backend-focused Full Stack Engineer building production-grade platforms with React frontends, Node.js and Django backends, stateless authentication, async worker pipelines, Redis queues, and AI-powered services. Strong in both UI architecture and distributed systems design with hands-on experience in production observability, security-first API design, and scalable microservice patterns.

CORE TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript
Frontend: React 18, Next.js 14, Vite, TanStack Query, Redux Toolkit, Framer Motion, React Hook Form, Tailwind CSS
Backend: Node.js, Express.js, Django, FastAPI, RESTful API Design, JWT Authentication, Zod, bcrypt
Real-Time & Distributed: WebSockets, Socket.IO, AsyncIO, Redis, Bull Queues, Celery, Event-Driven Architecture, Microservices
Databases: MongoDB, PostgreSQL, Redis, Mongoose ODM, Database Indexing, Query Optimization
DevOps & Observability: Docker, Prometheus, Sentry, Winston, OpenTelemetry, CI/CD, ESLint, Jest, Supertest, Git

KEY PROJECTS

Capsule — Secure AI-Moderated Media Platform
Live: capsule.topiiaa.site | github.com/Topiia/Capsule
Tech: Node.js, Express, MongoDB, Redis, Bull Queue, React, JWT, Prometheus, Sentry

- Engineered security-first backend architecture implementing stateless JWT authentication with HttpOnly cookie token delivery and refresh-token rotation with reuse detection.
- Designed Redis-backed Bull worker pipelines for AI moderation, email delivery, and asynchronous background processing to maintain low API latency under concurrent workloads.
- Built AI pre-moderation pipeline using Redis-backed worker queues to analyze uploads, generate content trust scores, and automatically flag unsafe media before publication.
- Implemented production observability stack using Prometheus metrics, Sentry error tracking, and structured logging for real-time monitoring and incident debugging.
- Built resilient async infrastructure including retry strategies, Dead-Letter Queue monitoring, and queue failure spike detection for background workers.
- Achieved full test stability with 162/162 passing unit and integration tests validating secure authentication flows and queue reliability.
- Designed stateless API architecture enabling horizontal scaling of services and background workers.
- Built full React 18 + Vite frontend with TanStack Query optimistic UI updates, 7-theme CSS variable engine, glassmorphism design system, and React.lazy code-split routing across all views.

CryptoDash — Real-Time Crypto Streaming Dashboard
Tech: React, Python, WebSockets, AsyncIO

- Built distributed real-time price streaming system using WebSockets and AsyncIO services.
- Implemented separate HTTP and WebSocket services to maintain consistent market state.
- Designed responsive charting dashboard with live connection health monitoring.

System Pulse — Real-Time System Monitoring Platform
Tech: Node.js, Redis, WebSockets, React, Prometheus

- Built real-time system monitoring platform streaming server metrics using WebSockets for live telemetry dashboards.
- Designed event-driven backend pipeline to collect and broadcast CPU, memory, and service health metrics.
- Built Redis-backed message channels for scalable metric streaming between monitoring agents and clients.

PROFESSIONAL EXPERIENCE

Python & Django Backend Intern
MIRA Advanced Engineering — Jan 2024 – Mar 2024

- Built backend modules and CRUD APIs using Django and Django REST Framework.
- Implemented REST endpoints and backend validation logic for internal applications.
- Assisted in API integration and backend data processing workflows.

EDUCATION

Bachelor of Engineering (Electronics & Telecommunication)
Terna Engineering College — 2021–2025
Higher Secondary Education (HSC) — 2021
Secondary (SSC) — 2019

CERTIFICATIONS

Python Programming, Django Development, Machine Learning, NLP Fundamentals